

Interfacial Structure and Tension of Polyelectrolyte Complex Coacervates

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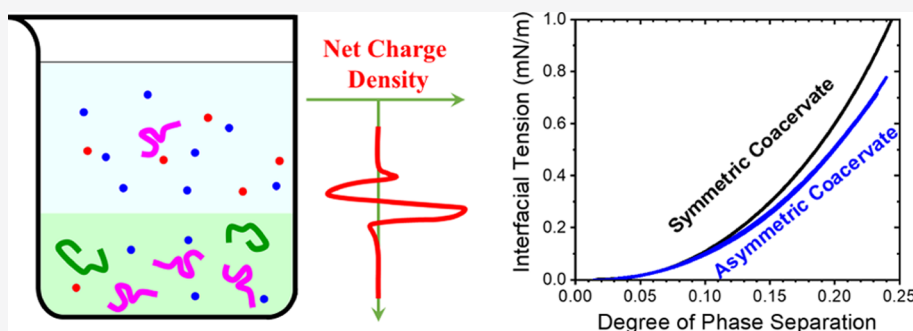
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ABSTRACT: We develop a simple inhomogeneous mean-field theory to study the interfacial structure and tension of polyelectrolyte complex coacervates in equilibrium with a supernatant solution. Our theory treats the electrostatic correlation by combining the Debye–Hückel theory with the first-order thermodynamic perturbation theory within the local density approximation, and incorporates the conformation entropy contribution for both polyions using Lifshitz’s ground-state dominance approximation. Using this theory, we systematically examine the interfacial properties of both symmetric and concentration-asymmetric coacervates. The interfacial tension γ is generally rather low, on the order of 1 mN/m or less. For asymmetric coacervates, an intricate electric double layer forms in the interfacial region, which can even contain several oscillations under certain conditions. The interfacial tension generally decreases with increasing the stoichiometric asymmetry, the added-salt concentration, and the initial polymer concentration of the mixture. We further find that the interfacial tension can be quantitatively related to the degree of phase separation S , where S is the Euclidean distance in composition between the two coexisting phases. In particular, we find that γ as a function of S for different concentration asymmetries collapses approximately to two master curves, which merge together and follow $\gamma \sim S^3$ for small S .

1. INTRODUCTION

Polyelectrolyte complex coacervation refers to an associative liquid–liquid phase separation in solutions of oppositely charged polyelectrolytes into a polymer-rich liquid coacervate phase and an equilibrium polymer-deficient supernatant phase.^{1–5} Due to its fundamental role in a variety of fields such as hydrogel fabrication,^{6–9} drug delivery,^{10,11} and formation of membrane-less organelles in cells,^{12–17} polyelectrolyte complex coacervation has received a great deal of attention in recent years from both the polymer physics and biophysics communities, and significant progresses have been made in understanding the bulk phase behaviors.^{2,18–50} For example, it is now well accepted that the addition of monovalent salt can weaken the degree of phase separation, and the miscibility gap vanishes above a critical salt concentration.^{21,23} Likewise, the effects of other parameters such as temperature, charge fraction, chain length, polymer concentration, sequence, and stoichiometry on the bulk properties of complex coacervation have been extensively

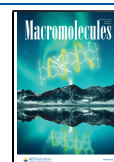
examined in the literature; we refer to several recent excellent reviews^{5,51–55} for these progresses.

Interfacial tension is a fundamental property of phase-separated systems. For polyelectrolyte complex coacervates, several experimental and theoretical studies^{40,56–65} have been performed to examine the interfacial tension of symmetric mixtures; these progresses have been summarized in a recent review by Prabhu.⁶⁶ Specifically, using a colloidal probe atomic force microscope, Spruijt et al.⁵⁶ found that the interfacial tension γ of symmetric coacervates decreases with increasing salt concentration ϕ_s and obeys a scaling relation $\gamma \sim |\phi_{s,c} - \phi_s|^{3/2}$ near the critical salt concentration $\phi_{s,c}$. This salt dependence has also been substantiated by Priftis et al.⁵⁹ using

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the surface force apparatus measurement and by Ali and Prabhu⁶⁴ using the deformed droplet retraction technique. Theoretically, this salt scaling relation has been rationalized by Qin and co-workers⁶⁰ who combined the Voorn–Overbeek theory² with the Cahn–Hilliard theory,⁶⁷ by Riggleman et al.⁵⁸ using the field theoretic simulation, and by Lytle et al.⁶³ using the transfer matrix method. Moreover, using a scaling analysis and a random-phase approximation (RPA) approach, Rumyantsev et al. examined the effects of solvent quality and salt concentration on the bulk behavior and interfacial tension in weakly charged symmetric coacervates.⁴⁰ Very recently, by combining experimental measurements and molecular dynamics simulations, Audus et al. found that $\gamma = \gamma_\infty(1 - h/N)$ far from the critical point, with γ_∞ being the interfacial tension in the long-chain limit, N being the degree of polymerization of polyions, and h being a constant.⁶⁵

All these studies, however, focus on the symmetric coacervates where polycations and polyanions have the same parameters such as chain length, charge fraction, and concentration. To the best of our knowledge, only Wang et al.⁶⁸ briefly examined the interfacial tension of the asymmetric coacervates using a classical density functional theory. However, these authors assumed the osmotic pressure to be fixed, and thus, their results cannot be directly compared with those of the experiments. A systematic investigation on how asymmetry affects the interfacial tension of complex coacervates is still lacking. Recently, using a simple liquid-state theory,⁴³ we studied the concentration-asymmetric coacervates and found a variety of novel phase separation behaviors different from those of the symmetric coacervates. One noteworthy observation is that the concentrations of both polyions are different in each of the two phases, and this concentration asymmetry leads to a finite Galvani potential between the two coexisting phases. This finite Galvani potential implies the formation of an electric double layer at the interface between the two coexisting phases, which is rather different from the symmetric coacervates, for which the interface is locally charge-neutral everywhere and the Galvani potential is identically zero. It is then natural to expect that the interfacial structure and tension of asymmetric coacervates should be different from those of the symmetric coacervates, and we thus aim to systematically examine the interfacial properties of asymmetric coacervates in this work.

To this end, we develop a simple inhomogeneous mean-field theory of polyelectrolyte complex coacervates. Our theory treats the electrostatic correlation by combining the size-corrected Debye–Hückel theory⁶⁹ with the first-order thermodynamic perturbation theory^{70–72} within the local density approximation, and incorporates the conformation entropy contribution for both polyions using Lifshitz's ground-state dominance approximation.^{73,74} Using this theory, we systematically examine the interfacial properties of both symmetric and concentration-asymmetric coacervates. We find that the interfacial tension γ is generally rather low and is on the order of 1 mN/m or less, qualitatively consistent with recent experimental observations.^{56,59,64} For symmetric coacervates, γ follows the expected “3/2” mean-field power-law behavior when approaching the critical salt concentration. For asymmetric coacervates, we find that an intricate electric double layer forms in the interfacial region, which can even include several oscillations under certain conditions. The resulting interfacial tension generally decreases with increasing the asymmetry of the solution, the added-salt concentration,

and the initial polymer concentration. We further find that the interfacial tension can be quantitatively related to the degree of phase separation.

The rest of this article is organized as follows. In the next section, we first develop a mean-field theory for the homogeneous polyelectrolyte mixture. In particular, we derive a new free energy expression due to the chain connectivity based on the Debye–Hückel theory for a simple electrolyte solution. This theory takes into account the electrostatic correlation between the nearest neighboring charged monomers along the chain backbone even if they are mediated by neutral monomers and thus goes beyond the traditional first-order thermodynamic perturbation theory. We then generalize this theory to the inhomogeneous polyelectrolyte solution. In Section 3, we use this theory to examine the interfacial structure and tension of polyelectrolyte complex coacervates, with a focus on the asymmetric coacervates. Finally, a summary of the key results along with a brief outlook is given in Section 4.

2. MODEL AND THEORY

2.1. Model. We consider a strongly charged polyelectrolyte solution having n_{p+} polycations and n_{p-} polyanions in a system of volume V . Each polycation (polyanion) consists of a total of N_{p+} (N_{p-}) coarse-grained monomers connected consecutively via a bonding potential $\Phi(\mathbf{r})$. To explicitly incorporate the charge sequence along the chain backbone, we introduce a parameter $\alpha_{i,s}$ to represent the type of the s th monomer on the backbone of the i th species. Specifically, $\alpha = 1$ and -1 denote, respectively, a positively and negatively charged monomer, and $\alpha = 0$ denotes a neutral monomer. Therefore, the number of charged monomers on each polyion of species i is $N_{c,i} \equiv \sum_{s=1}^{N_i} |\alpha_{i,s}|$, and the corresponding charge fraction of species i is $f_i = N_{c,i}/N_i$. Furthermore, we denote the index of the k th charged monomers along the chain backbone in species i by $j_{i,k}$ with k running from 1 to $N_{c,i}$. For simplicity, we further assume that the counterions dissociated from polyions are the same as the salt ions. Representing the numbers of cations, anions, and solvents as n_+ , n_- , and n_s , respectively, and assuming all monomers, ions, and solvents to have the same volume v , we have $V = (n_{p+}N_{p+} + n_{p-}N_{p-} + n_+ + n_- + n_s)v$. Generalization to account for the size disparity among different species is straightforward. The volume fraction of the species i is then given by $\phi_i = n_i N_i v / V$ with $N_+ = N_- = N_s = 1$. Finally, we denote the valency of a polycation monomer, polyanion monomer, cation, and anion by z_{p+} , z_{p-} , z_+ , and z_- , respectively.

2.2. Homogeneous Solution. For a homogeneous bulk solution, we may write the Helmholtz free energy density as

$$f_b = f_{\text{FH}} + f_{\text{el}} + f_{\text{ch}} \quad (1)$$

The first contribution in eq 1 is the Flory–Huggins free energy density as given by

$$\begin{aligned} \frac{v f_{\text{FH}}}{k_B T} = & \sum_{i=p\pm, \pm} \frac{\phi_i}{N_i} \ln \phi_i + \left(1 - \sum_{i=p\pm, \pm} \phi_i \right) \ln \left(1 - \sum_{i=p\pm, \pm} \phi_i \right) \\ & + \sum_{i,j \neq i} \chi_{ij} \phi_i \phi_j \end{aligned} \quad (2)$$

where k_B and T are the Boltzmann constant and the absolute temperature, respectively. The first two terms in eq 2 account for the mixing entropy of all species, and the last term

represents the nonelectrostatic interactions among all species with χ_{ij} being the Flory–Huggins interaction parameter between species i and j ; the summation in the third term is for all species including the solvent. While these χ_{ij} interactions definitely play an important role in determining the phase behavior,^{44,75–77} in this work, we set $\chi_{ij} = 0$ to highlight the role of the electrostatic interaction.

The second contribution in eq 2, f_{el} , arises from the electrostatic correlation of disconnected ions at the Gaussian fluctuation level [i.e., at the Debye–Hückel (DH) level].⁶⁹ While the point charge form of the DH free energy is commonly used in the Voorn–Overbeek theory, here we adopt the expression that takes into account the finite size of ions,^{78,79} and the corresponding expression is given by

$$\frac{v f_{\text{el}}}{k_{\text{B}} T} = -\frac{1}{4\pi} \left[\ln(1 + \kappa\sigma) - \kappa\sigma + \frac{(\kappa\sigma)^2}{2} \right] \quad (3)$$

where the inverse Debye screening length κ is given by $\kappa\sigma = \sqrt{4\pi(l_{\text{B}}/\sigma) \sum_{i=p\pm, \pm} f_i \phi_i z_i^2}$, with the Bjerrum length $l_{\text{B}} \equiv e^2/4\pi k_{\text{B}} T \epsilon_0 \epsilon_r$ (where e is the elementary charge, ϵ_0 is the vacuum permittivity, and ϵ_r is the dielectric constant of the medium), $\sigma \equiv v^{1/3}$ being the size of the species. In the dilute limit, $\phi_i \rightarrow 0$, eq 3 reduces to the well-known expression for point charge, $f_{\text{DH}} = -\kappa^3/12\pi$ upon setting $v = \sigma^3$. Here, we use the size-corrected free energy, eq 3, as the excluded volume effects are known to weaken the electrostatic correlation at higher concentrations,⁷⁹ and by including the ion sizes, this expression captures some of the excluded volume effects on electrostatic correlations for higher electrolyte concentrations compared to the point charge DH limiting expression. We note that this size-corrected DH free energy density is commonly used in the literature^{33,39,80,81} to study the thermodynamics of polyelectrolyte complex coacervation.

The last term in eq 1 accounts for the chain connectivity contribution to the electrostatic correlation due to the fact that some of the ions included in the DH theory are from connected polyelectrolyte chains. The Voorn–Overbeek theory,² missing the chain connectivity contribution in the electrostatic correlation, results in an incorrect description on the salt partition between the supernatant phase and coacervate phase.^{23,31,37,38,42} Moreover, the chain connectivity also plays an important role in determining the phase boundary and thus the polyion concentrations in the coexisting phases.^{32,42} Several advanced theories, such as the liquid-state theory,^{42,43} the renormalized Gaussian fluctuation theory,^{24,82} and the transition matrix theory,³⁷ have been developed to account for the chain connectivity effect. Here, we approximate F_{ch} in the framework of the first-order thermodynamic perturbation theory (TPT1), originally developed by Wertheim^{70,71} for the tangentially connected hard sphere chain with a fixed bond length and later generalized by MacDowell et al. to models with variable bond lengths.⁷² Specifically, F_{ch} can be approximated via the radial distribution function of a simple electrolyte solution g_{ii} between the same small ionic species i and the bonding potential $\Phi(\mathbf{r})$, that is

$$F_{\text{ch}} = -k_{\text{B}} T \sum_{i=p+, p-} n_i (N_i - 1) \ln \left\{ \int d\mathbf{r} g_{ii}(\mathbf{r}) \exp[-\Phi(\mathbf{r})/k_{\text{B}} T] \right\} \quad (4)$$

where $N_i - 1$ refers to the number of bonds per chain of species i . For the tangentially connected hard-sphere chain model where the bond length is the same as the hard sphere diameter, the integral in eq 4 can be replaced by the radial distribution function at contact $g_{ii}(\sigma)$ that is analytically known under the mean-spherical approximation,^{83–85} and the corresponding excess free energy becomes

$$F_{\text{ch}} = -k_{\text{B}} T \sum_{i=p+, p-} n_i (N_i - 1) \ln g_{ii}(\sigma) \quad (5)$$

where an inconsequential constant has been ignored. This free energy has been widely used to investigate the thermodynamics and phase behaviors of neutral polymer solutions^{72,86,87} and polyelectrolyte solutions.^{43,88–90} We refer to our recent work^{43,90} for the details of this theoretical framework.

For the partially charged cases, however, this excess free energy cannot capture the electrostatic correlation due to chain connectivity. For example, for the specific case with alternating charged and neutral monomers along the chain backbone, the electrostatic correlation contribution due to chain connectivity is completely missed because the $g_{ii}(\sigma)$ will only include the nonelectrostatic contribution. This will result in significant underestimation of the electrostatic correlation and thus affect the thermodynamics and phase behavior of a polyelectrolyte solution. To improve this shortcoming, we generalize the TPT1 theory to incorporate the electrostatic correlation between neighboring charged residues along the chain backbone. Specifically, as mentioned above, we denote the total number of charged monomers per chain in species i as $N_{c,i}$ and the index of the k th charged monomers along the chain backbone in species i as $j_{i,k}$. To pursue an analytical expression, we assume that the electrostatic correlation does not significantly affect the chain conformation, and the average distance between the k th and $k+1$ th charged monomers can be approximated by the internal distance between segments in a Gaussian chain, that is, $r_{k,k+1} = (j_{i,k+1} - j_{i,k})^{1/2} \sigma$. Neutron/X-ray/light scattering experiments by Spruijt et al. showed that both polycations and polyanions in symmetric coacervates have ideal Gaussian chain conformation,⁹¹ which partially justifies our assumption here. We note that most analytical theories on polyelectrolyte complex coacervation—Voorn–Overbeek,² generalized Voorn–Overbeek,³³ PRISM,³¹ and RPA^{32,92}—implicitly or explicitly assume ideal chain conformation. In a similar spirit to eq 5, we may write

$$F_{\text{ch}} = -k_{\text{B}} T \sum_{i=p+, p-} n_i \left[\sum_{k=1}^{N_{c,i}-1} \ln g_{ii}((j_{i,k+1} - j_{i,k})^{1/2} \sigma) \right] \quad (6)$$

In the framework of the Debye–Hückel theory, the radial distribution function of an electrolyte solution with the diameter of all species σ is given by^{78,79}

$$g_{ii}(r) = 1 - \frac{z_i^2 l_{\text{B}}}{(1 + \kappa\sigma)r} \exp[-\kappa(r - \sigma)] \quad (7)$$

for $r > \sigma$ and 0 otherwise. To proceed, we assume that the second contribution of eq 7 is much smaller than 1 and then approximate the logarithm with the leading term in the Taylor expansion. The resulting free energy density due to chain connectivity is thus

$$\frac{v f_{\text{ch}}}{k_{\text{B}} T} = \frac{v F_{\text{ch}}}{k_{\text{B}} T V} = k_{\text{B}} T \sum_{i=p+, p-} \frac{l_{\text{B}} \phi_i}{N_i} \left[\sum_{k=1}^{N_{i,i}-1} z_i^2 \frac{\exp[-\kappa((j_{i,k+1} - j_{i,k})^{1/2} - 1)\sigma]}{(1 + \kappa\sigma)(j_{i,k+1} - j_{i,k})^{1/2}\sigma} \right] \quad (8)$$

We note that the TPT1 theory developed here also only includes the electrostatic correlations between neighboring charged monomers along the backbone (which can still have several monomers apart) and thus neglects the correlation beyond the neighboring charged monomers. For fully charged polyelectrolyte chains, it has been shown that TPT1 underestimates the electrostatic correlation due to chain connectivity.⁸² Nevertheless, the qualitative consistency of our theoretical phase diagram with the experimental one (see the Results and Discussion section) suggests that this TPT1 theory is able to capture the key physical factors governing the phase behaviors of complex coacervation. We expect the accuracy of TPT1 to improve with more sparsely charged chains, although in that case, the chain connectivity effect on the electrostatic correlation becomes less important.

For the fully charged model, $N_{i,i} = N_i$ and $j_{i,k+1} - j_{i,k} = 1$ (i.e., $\alpha_{p+} = 1$ and $\alpha_{p-} = -1$ for all monomers); eq 8 then reduces to

$$\frac{v f_{\text{ch}}}{k_{\text{B}} T} = - \sum_{i=p+, p-} \frac{\phi_i z_i^2 l_{\text{B}} / \sigma}{1 + \kappa\sigma} (N_i^{-1} - 1) \quad (9)$$

It is worth noting that eqs 3, 8, and 9 are derived based on the DH treatment of the electrostatic effects, and are thus only applicable to systems with weak electrostatic correlations. For systems with strong electrostatic correlations, effects such as ion pairing need to be explicitly included to correctly describe the structure and thermodynamics of the coacervates. However, the consideration of these effects is beyond the scope of this work. In this work, we limit the choice of our parameters to stay within the regimes of weak and moderate electrostatic correlations. Hereafter, we focus on the fully charged polyelectrolyte for simplicity; the study of the partially charged polyelectrolyte will be deferred to the future. Combining all the three contributions given by eqs 2, 3, and 9, we obtain the dimensionless Helmholtz free energy density of a bulk homogeneous polyelectrolyte solution as

$$\begin{aligned} \frac{v f_{\text{b}}}{k_{\text{B}} T} = & \sum_{i=p\pm, \pm} \frac{\phi_i}{N_i} \ln \phi_i + \left(1 - \sum_{i=p\pm, \pm} \phi_i \right) \ln \left(1 - \sum_{i=p\pm, \pm} \phi_i \right) \\ & - \frac{1}{4\pi} \left[\ln(1 + \kappa\sigma) - \kappa\sigma + \frac{(\kappa\sigma)^2}{2} \right] \\ & - \frac{l_{\text{B}} / \sigma}{1 + \kappa\sigma} \sum_{i=p+, p-} (N_i^{-1} - 1) (z_i^2 \phi_i) \end{aligned} \quad (10)$$

We then calculate the bulk chemical potential of species i as $\mu_{b,i} = [\partial(f_{\text{b}} v) / \partial \phi_i]_{\phi_j \neq \phi_i}$ (for polyions, the chemical potential is on a per monomer basis) and the osmotic pressure as $\Pi_{\text{b}} = v^{-1} \sum_{i=p\pm, \pm} \mu_{b,i} \phi_i - f_{\text{b}} + f_{\text{b}}(\{\phi_i = 0\})$. Detailed expressions for the chemical potentials and osmotic pressure are given in the Supporting Information. The concentrations of all species in the two coexisting phases are determined by the equality of the

osmotic pressure and the electrochemical potential for all species in the two coexisting phases. Denoting $\{\phi_i^{\text{I}}\}$ and $\{\phi_i^{\text{II}}\}$ as the concentration of species i in the supernatant phase and the coacervate phase, respectively, we thus have the electrochemical potential of species i in the supernatant phase (i.e., phase I) and in the coacervate phase (i.e., phase II), respectively, as $\mu_{b,i}(\{\phi_i^{\text{I}}\}) + e z_i \psi^{\text{I}}$ and $\mu_{b,i}(\{\phi_i^{\text{II}}\}) + e z_i \psi^{\text{II}}$, where ψ^{I} and ψ^{II} are the mean electrostatic potential in these two phases, respectively. The condition of phase equilibrium requires $\Pi_{\text{b}}(\{\phi_i^{\text{I}}\}) = \Pi_{\text{b}}(\{\phi_i^{\text{II}}\})$ and $\mu_{b,i}(\{\phi_i^{\text{I}}\}) = \mu_{b,i}(\{\phi_i^{\text{II}}\}) + e z_i \Psi_{\text{G}}$, where the Galvani potential $\Psi_{\text{G}} \equiv \psi^{\text{II}} - \psi^{\text{I}}$ represents the electrostatic potential difference between the coacervate phase and the coexisting supernatant phase. Physically, the Galvani potential Ψ_{G} arises from the requirement of the equality of electrochemical potential for each species and the charge neutrality in each of the coexisting phases. For symmetric coacervates, because the properties of oppositely charged species are otherwise the same except for the opposite charges, charge neutrality is automatically satisfied, and thus, $\Psi_{\text{G}} = 0$. For asymmetric coacervates, the opposite charges on the polyions do not automatically compensate each other by themselves, and extra small ions are required to maintain charge neutrality; a finite Ψ_{G} is thus required to meet the conditions of the equality of electrochemical potential for each species in the coexisting phases. At a given l_{B} , ϕ_i^{I} and ϕ_i^{II} for all species along with Ψ_{G} are then calculated by numerically solving these phase equilibrium equations and charge neutrality constraints using the Newton–Raphson method. The general procedures to construct the bulk phase diagram have been presented in our previous work in detail⁴³ and thus will not be repeated here.

2.3. Inhomogeneous System. For an inhomogeneous system, the concentration of each species depends on the spatial position, and this inhomogeneity leads to a conformation entropy penalty for the polymer chains. Furthermore, local charge neutrality may be violated. To incorporate these two effects, we construct a minimal model that accounts for the polymer conformation entropy using the mean-field theory under the ground-state dominance approximation (GSDA) and the electrostatic interactions in terms of a generalized Poisson–Boltzmann theory. The Helmholtz free energy functional is

$$F = \int d\mathbf{r} \left[f_{\text{b}}(\{\phi_i(\mathbf{r})\}) + k_{\text{B}} T \sum_{i=p\pm} \frac{b^2 |\nabla \phi_i(\mathbf{r})|^2}{24 v \phi_i(\mathbf{r})} + \rho_{\text{e}}(\mathbf{r}) \psi(\mathbf{r}) - \frac{1}{2} \epsilon_0 \epsilon(\{\phi_i(\mathbf{r})\}) |\nabla \psi(\mathbf{r})|^2 \right] \quad (11)$$

where $f_{\text{b}}(\{\phi_i\})$ takes the bulk expression in eq 10 with ϕ_i here replaced by the local value $\phi_i(\mathbf{r})$. Note that similar local density approximations have been widely used in the literature, such as in the study of hydrogel formation⁹³ of charged triblock copolymer solution and the image charge effect in a confined simple electrolyte solution.⁹⁴ The second term is from the well-known Lifshitz entropy contribution that can be derived from the mean-field theory under the GSDA.^{73,74,95} The last two terms are the variational electrostatic free energy with $\rho_{\text{e}}(\mathbf{r}) \equiv e \sum_{i=p\pm, \pm} z_i \phi_i(\mathbf{r}) / v$ being the local charge density and $\psi(\mathbf{r})$ being the variational electrostatic potential. In general, the dielectric constant $\epsilon(\{\phi_i(\mathbf{r})\})$ depends on the local concentration of each species. However, in this work, we ignore this dependence to

highlight the role of electrostatic correlation. We note that our theory reduces to the theory by Qin et al.⁶⁰ for symmetric systems, if we ignore the chain connectivity contribution and write the electrostatic correlation using the point-like DH expression.

It is convenient to introduce $\varphi_i \equiv \sqrt{\phi_i}$ for the polymer species $i = p+, p-$. Re-expressing the inhomogeneous free energy functional in eq 11 using φ_i , we have

$$F = \int d\mathbf{r} \left[f_b(\{\phi_i(\mathbf{r})\}) + k_B T \sum_{i=p\pm} \frac{b^2}{6\nu} |\nabla \varphi_i(\mathbf{r})|^2 + \rho_e(\mathbf{r})\psi(\mathbf{r}) - \frac{1}{2} \varepsilon_0 \varepsilon |\nabla \psi(\mathbf{r})|^2 \right] \quad (12)$$

Extremizing the above free energy with respect to $\psi(\mathbf{r})$, we obtain the Poisson equation

$$\varepsilon_0 \varepsilon \nabla^2 \psi(\mathbf{r}) = -\rho_e(\mathbf{r}) \quad (13)$$

Minimizing F with respect to $\phi_j(\mathbf{r})$ under the constraints of a fixed amount of each species $\int d\mathbf{r} \phi_j(\mathbf{r}) = \bar{\phi}_j V$ and the charge neutrality $\sum_j z_j \int d\mathbf{r} \phi_j(\mathbf{r}) = 0$, we obtain the Euler–Lagrange equation

$$\mu_{b,j} + ez_j \psi_b = \frac{\partial [v f_b(\phi_j(\mathbf{r}))]}{\partial \phi_j} - \frac{k_B T b^2}{6} \sum_{i=p\pm} \frac{\nabla_r^2 \varphi_i}{\varphi_i} \delta_{i,j} + ez_j \psi(\mathbf{r}) \quad (14)$$

for both polyions (i.e., $j = p+, p-$) and salt ions (i.e., $j = +, -$), where $\mu_{b,j}$ and ψ_b are the Lagrange multipliers enforcing these two constraints. The equilibrium condition actually requires $\mu_{b,j}$ and ψ_b to be, respectively, the bulk chemical potential of species j and the bulk electrostatic potential. Physically, eq 14 is a statement of the equality of the local electrochemical potential at all $\mathbf{r}$ for all species.

For general interfacial problems, we need appropriate boundary conditions, depending on the nature of the boundary. Hereafter, we focus on the interfacial properties of a polyelectrolyte complex coacervate in equilibrium with a supernatant solution.

2.4. Interfacial Properties between the Two Coexisting Phases. As alluded to above, for an *asymmetric* complex coacervate, while charge neutrality is maintained in each of the coexisting phases, the asymmetry between the oppositely charged species gives rise to a finite Galvani potential. This Galvani potential indicates the formation of an electric double layer at the interface between the two coexisting phases. The equilibrium density profile and the mean electrostatic potential are obtained by solving eq 14 (for each species) and eq 13 simultaneously. For simplicity, here we ignore the capillary wave fluctuations and assume that the interface is flat and homogeneous in the parallel directions. Let x be the coordinate normal to the interface; the Poisson equation, eq 13, reduces to

$$\frac{\partial^2}{\partial x^2} [\beta e \psi(x)] = -\frac{4\pi l_B}{\nu} \sum_{i=p\pm, \pm} z_i \phi_i(x) \quad (15)$$

and eq 14 reduces to

$$b^2 \frac{\partial^2 \varphi_{p\pm}(x)}{\partial x^2} = \frac{6\varphi_{p\pm}}{k_B T} \left[\frac{\partial [v f_b(\{\phi_i(x)\})]}{\partial \phi_{p\pm}} - \mu_{b,p\pm} + ez_{p\pm} [\psi(x) - \psi_b] \right] \quad (16)$$

for polycations and polyanions (i.e., $p+, p-$), and

$$\frac{\partial [v f_b(\{\phi_i(x)\})]}{\partial \phi_{\pm}} - \mu_{b,\pm} + ez_{\pm} [\psi(x) - \psi_b] = 0 \quad (17)$$

for salt cations and anions (i.e., $+, -$).

For the case of two-phase coexistence, we may choose either the supernatant or the coacervate phase as the bulk reference state. For concreteness, we choose the supernatant phase as the reference and set $\psi_b^I = 0$ in eqs 16 and 17; the electrostatic potential of the coacervate phase ψ_b^{II} is then the Galvani potential Ψ_G . The value of the chemical potential of the species in the supernatant phase is determined by the bulk phase coexistence condition.

The variables $\psi(x)$, $\varphi_{p+}(x)$, $\varphi_{p-}(x)$, $\phi_+(x)$, and $\phi_-(x)$ are then calculated by numerically solving eqs 15–17 along with the boundary conditions

$$\psi(x = -\infty) = 0, \quad \psi(x = \infty) = \Psi_G \quad (18)$$

$$\varphi_j(x = -\infty) = \sqrt{\phi_j^I}, \quad \varphi_j(x = \infty) = \sqrt{\phi_j^{II}} \quad (19)$$

for $j = p+, p-$, where we have tacitly assumed that the supernatant phase occupies the left side of the space. It is worth noting that the position of the interface is not uniquely fixed by specifying the electrostatic chemical potential of the supernatant phase. We choose the origin of the coordinate system by further setting $\phi_{p+}(x=0) + \phi_{p-}(x=0) = (\phi_{p+}^I + \phi_{p-}^I + \phi_{p+}^{II} + \phi_{p-}^{II})/2$; this uniquely determines the location of the interface.

Once the density profiles of all species and the electrostatic potential are obtained, we further calculate the interfacial tension by the difference in the grand potential between the inhomogeneous system with the interface and the same system without the interface, divided by the interface area A . Noting that the grand free energy density of a bulk phase is the negative of the osmotic pressure $-\Pi_b$ and that at coexistence, the osmotic pressure is the same for the supernatant and the coacervate phases, the interfacial tension is

$$\gamma = \frac{1}{A} \left[F(\{\phi_j, \psi\}) - \frac{1}{\nu} \sum_{j=p\pm, \pm} (\mu_{j,b} + ez_j \psi_b) \int d\mathbf{r} \phi_j(\mathbf{r}) + \Pi_b V \right] \\ = \int_{-\infty}^{\infty} dx \left[f_b(\{\phi_i(x)\}) - \sum_{i=p\pm, \pm} \phi_i(x) \frac{\partial f_b}{\partial \phi_i} + \Pi_b \right] \\ - \frac{k_B T}{8\pi l_B} \int_{-\infty}^{\infty} dx \left(\frac{\partial [\beta e \psi(x)]}{\partial x} \right)^2 \quad (20)$$

From the second line of eq 20, we see that γ consists of two parts: The first part, henceforth denoted by γ_{ne} , depends only on the density profile of the species. The second part, henceforth denoted by γ_{el} , results from the formation of the electric double layer and is strictly negative.

For the special case of *symmetric* complex coacervation, there is no charge separation, and thus, the Poisson equation is

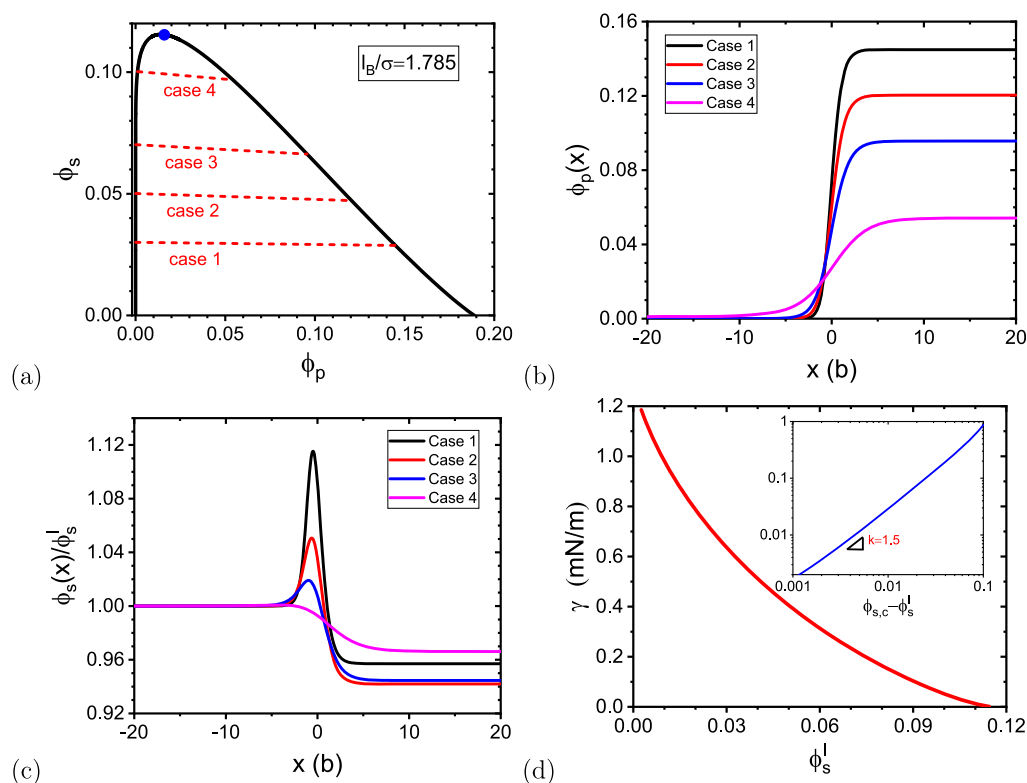


Figure 1. (a) Bulk phase diagram in the ϕ_p – ϕ_s plane. The black curve is the binodal, and the dashed lines denote several representative tie lines. The density profile for the polymer segments (b) and density profile for the salt ions (c) for the four representative tie lines specified in (a). (d) Interfacial tension as a function of the volume fraction of salt in the supernatant phase ϕ_s^I . The inset shows the 1.5 scaling exponent close to the critical salt concentration.

irrelevant. Further denoting $\phi_p = \phi_{p\pm}$, $\varphi_p = \varphi_{p\pm}$, and $\phi_s = \phi_{\pm}$, eqs 16 and 17 reduce to

$$b^2 \frac{\partial^2 \varphi_p(x)}{\partial x^2} = \frac{6\varphi_p(x)}{k_B T} [\mu_p(x) - \mu_{b,p}] \quad (21)$$

$$\mu_s(x) = \mu_{b,s} \quad (22)$$

respectively, where $\mu_i(x) \equiv \partial[vf_b(\{\phi_i(x)\})]/\partial\phi_i$ is the local chemical potential of species i . The boundary conditions for the second-order ordinary differential eq 21 are $\phi_{p\pm}(x = -\infty) = \phi_{p\pm}^I$ and $\phi_{p\pm}(x = \infty) = \phi_{p\pm}^{II}$.

For the numerical results presented below, we assume that both species of polyions are fully charged and then set $N_{p+} = N_{p-} = 100$ and $z_{p+} = z_+ = -z_{p-} = -z_- = 1$ for simplicity. Moreover, we set the size of all species $\sigma = 0.4$ nm; thus, at the room temperature $T = 300$ K, an aqueous solution corresponds to $l_B/\sigma = 1.785$. In addition, we fix $b = \sigma$ to minimize the number of parameters.

3. RESULTS AND DISCUSSION

In this section, we first examine the interfacial properties of the symmetric coacervate, in which the oppositely charged species have identical concentrations, and compare our results with those in the literature. We then discuss the concentration-asymmetric case. By concentration asymmetry, we mean the stoichiometric asymmetry between the polycation solution and the polyanion solution when preparing the mixture. Both polyelectrolyte solutions have the same concentration and otherwise the same parameters, such as the chain length, charge fraction, and salt concentration, but the final mixture is

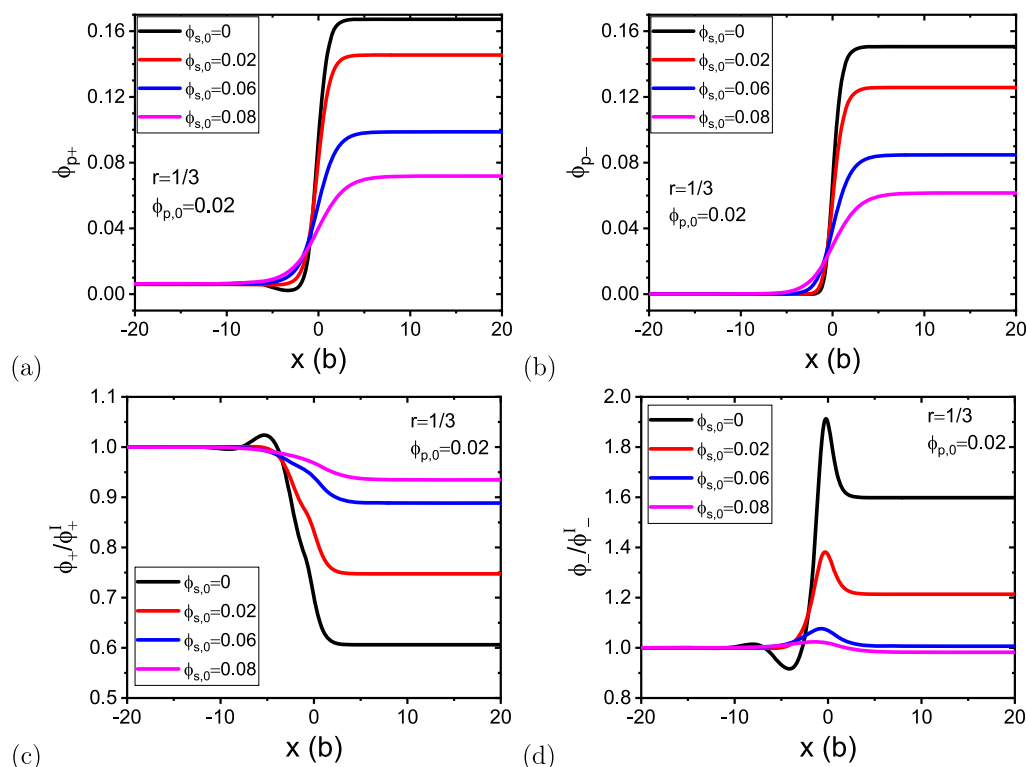
made from mixing a volume V_1 of the polycation solution with a volume V_2 of the polyanion solution. The bulk phase behavior of this mixture has been systematically examined in our previous work using a simple liquid-state theory.⁴³ Similar results are obtained using the current theory developed in Section 2 and therefore will not be presented; we refer the readers to the Supporting Information for some representative results. In this work, we focus on the interfacial properties of the asymmetric coacervate.

Let $\phi_{p,0}$ and $\phi_{s,0}$ denote the concentration of polyions and salt, respectively, in the initial polycation and polyanion solutions. The overall concentration of polycations, polyanions, cations, and anions in the mixture is thus $\bar{\phi}_{p+} = (1+r)\phi_{p,0}/2$, $\bar{\phi}_{p-} = (1-r)\phi_{p,0}/2$, $\bar{\phi}_+ = (1-r)\phi_{p,0} + \phi_{s,0}$, and $\bar{\phi}_- = (1+r)\phi_{p,0} + \phi_{s,0}$, respectively, where the asymmetry factor $r \equiv (V_1 - V_2)/(V_1 + V_2)$ characterizes the stoichiometric asymmetry when preparing the mixture. Specifically, $r = 0$ corresponds to a symmetric mixture, and positive r corresponds to an excess of polycations over polyanions. For example, $r = 1/3$ and $r = 1/2$ means, respectively, that the volume of the polycation solution is twice and three times the volume of the polyanion solution when preparing the mixture. The phase behavior and the interfacial properties of this mixture are then uniquely determined by the following three parameters, $\phi_{s,0}$, $\phi_{p,0}$, and r , that are easily controlled in experiments.

3.1. Symmetric Complex Coacervation and Interfacial Structures. For the symmetric mixture, the polycation and the polyanion have the same concentration in each of the coexisting phases, and the bulk phase behaviors of this system have been extensively studied in the literature.^{21,23,30} We thus first examine whether our theory presented in Section 2 can

Table 1. Bulk Volume Fraction of Each Species and the Galvani Potential at Coexistence for Several Representative Values of Added-Salt Concentration $\phi_{s,0}$ with Fixed $r = 1/3$ and $\phi_{p,0} = 0.02$

$\phi_{s,0}$	ϕ_{p+}^I	ϕ_{p-}^I	ϕ_+^I	ϕ_-^I	ϕ_{p+}^{II}	ϕ_{p-}^{II}	ϕ_+^{II}	ϕ_-^{II}	$\beta e \Psi_G$
0	0.00620	0	0.00678	0.0130	0.167	0.151	0.00411	0.0208	0.485
0.02	0.00593	0	0.0270	0.0330	0.146	0.126	0.0202	0.0400	0.242
0.06	0.00604	0	0.0673	0.0733	0.0987	0.0847	0.0598	0.0738	0.0624
0.08	0.00627	0	0.0873	0.0935	0.0718	0.0616	0.0816	0.0919	0.0249

**Figure 2.** Density profile of (a) polycation, (b) polyanion, (c) small cation, and (d) small anion, for asymmetric coacervates with $r = 1/3$ and $\phi_{p,0} = 0.02$ fixed. In panels (c,d) the vertical axis is normalized by the bulk volume fraction of cations and anions in the supernatant phase, respectively.

correctly describe the bulk phase behavior. In Figure 1a, we show the binodal (i.e., the black curve) in the surface of polymer–salt concentrations. The red dash curves denote several representative tie lines that connect the concentrations of each species in the coexisting phases, and the blue solid circle is the upper critical point. We see that this simple theory indeed qualitatively captures the main features of coacervation in symmetric mixtures for strongly charged polyelectrolyte solutions observed in the literature, that is, the existence of an upper critical salt concentration^{21,30} and the negative slope of the tie line.^{23,31,42} We note that the slope of the tie line can become positive for weakly charged polyelectrolytes when an effective attraction between polyions and salt ions is present.⁷⁷

We now examine the interfacial structure and tension of the symmetric coacervate. Since the mean concentrations of the positively charged species are the same as the counterparts of the negatively charged species everywhere, we define $\phi_p(x) \equiv \phi_{p+}(x) = \phi_{p-}(x)$ and $\phi_s(x) \equiv \phi_+(x) = \phi_-(x)$ for brevity. In Figure 1b,c, we show the polyion concentration $\phi_p(x)$ and salt concentration $\phi_s(x)$, respectively, in the direction normal to the interface between the coacervates and its coexisting supernatant phase, for the four representative tie lines specified in Figure 1a. For each case, $\phi_p(x)$ increases monotonically from the supernatant phase to the coacervate phase, and the width of the interface becomes larger with increasing the salt

concentration. On the other hand, while the magenta curve in Figure 1c shows that $\phi_s(x)$ decreases monotonically at a high salt concentration, the other three curves suggest that a slight enrichment of salt occurs at the interface when the salt concentration is low or modest. We note that the monotonic decrease in $\phi_s(x)$ is consistent with the results reported in previous studies;^{58,63} the nonmonotonic behavior of $\phi(x)$, however, has not been reported previously and remains to be verified experimentally or with computer simulation in the future. To understand this nonmonotonic behavior, we note that at small and modest salt concentrations, the interface is quite sharp, which incurs a large chain conformation entropy penalty. The enrichment of salt in the interface may help alleviate this conformation entropy penalty. With increasing salt concentration, the interface widens significantly, and the conformational entropy penalty is reduced and thus $\phi_s(x)$ varies monotonically across the interface.

Previous studies have shown that the interfacial tension of the liquid coacervates is rather low, that is, on the order of mN/m, depending on the specific system and the salt concentration. In Figure 1d, we show the interfacial tension γ as a function of the salt concentration in the supernatant phase ϕ_s^I . We see that γ is indeed on the order of 1 mN/m and decreases monotonically with ϕ_s^I . Furthermore, the inset of Figure 1d shows that $\gamma \sim (\phi_{s,c} - \phi_s^I)^{3/2}$, close to the critical salt

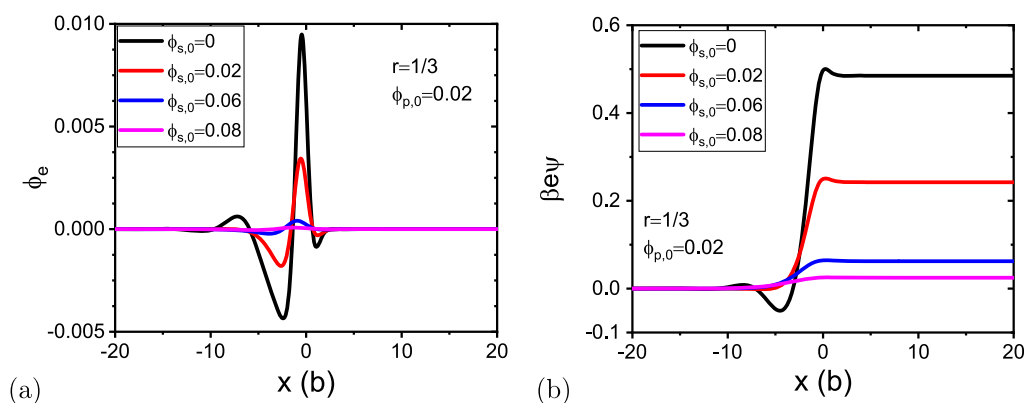


Figure 3. (a) Net charge density ϕ_e and (b) mean electrostatic potential ψ , for asymmetric coacervates with $r = 1/3$ and $\phi_{p,0} = 0.02$.

concentration $\phi_{s,c}$. The scaling exponent $3/2$ reflects the mean-field nature of our theory. These results are qualitatively consistent with previous experimental observations^{56,59,64} and theoretical calculations.^{58,60,63}

3.2. Interfacial Structure and Tension of Asymmetric Coacervates. In this section, we discuss the interfacial properties of asymmetric coacervates. In the [Supporting Information](#), we present several illustrating examples to show the dependence of the bulk properties on $\phi_{s,0}$, $\phi_{p,0}$, and r . Here, we briefly summarize the main results, since they provide the basis for understanding the interfacial properties. For concreteness, we assume that the polycations are the major polyanion species, and thus, $r > 0$. Accordingly, the small cations and anions are, respectively, the minor and major salt ion species. Different from the symmetric case, for moderate values of r , the polycation concentration in the supernatant phase ϕ_{p+}^I takes a moderate value, while $\phi_{p-}^I \approx 0$, even far from the critical point. In the coacervate phase, ϕ_{p-}^{II} is slightly less than ϕ_{p+}^{II} . More interestingly, due to the asymmetry between the oppositely charged species, a finite Galvani potential Ψ_G naturally emerges between the two coexisting phases, with the concurrent formation of an electric double layer in the interfacial region. In the following, we examine the interfacial structure and tension of asymmetric coacervates in detail.

3.3. Effects of Added-Salt Concentration $\phi_{s,0}$. First, we examine the effect of the concentration of added salt $\phi_{s,0}$ on the interfacial structure and tension for the asymmetric coacervate. At given r and $\phi_{p,0}$, the overall polycation and polyanion concentrations $\bar{\phi}_{p+} = (1 + r)\phi_{p,0}/2$ and $\bar{\phi}_{p-} = (1 - r)\phi_{p,0}/2$, respectively, remain unchanged; the overall salt ion concentrations, $\bar{\phi}_+ = (1 + r)\phi_{p,0}/2 + \phi_{s,0}$ and $\bar{\phi}_- = (1 - r)\phi_{p,0}/2 + \phi_{s,0}$, however, increase linearly with $\phi_{s,0}$. Taking the system with $r = 1/3$ and $\phi_{p,0} = 0.02$ as an example (corresponding to $\bar{\phi}_{p+}/\bar{\phi}_{p-} = 2$), in [Table 1](#), we present the concentrations of all species in both the supernatant phase (denoted by the superscript “I”) and the coacervate phase (denoted by the superscript “II”) and the Galvani potential Ψ_G , at several representative values of $\phi_{s,0}$. These quantities are used as the boundary conditions when solving the interfacial properties. We see that with increasing $\phi_{s,0}$, ϕ_{p+}^I remains nearly unchanged, but both $\phi_{p\pm}^{II}$ and Ψ_G decrease monotonically. The polyanion concentration in the supernatant phase ϕ_{p-}^I is many orders of magnitude smaller than the polycation concentration ϕ_{p+}^I and thus abbreviated as 0 in [Table 1](#). ϕ_{p+}^I , however, remains approximately as a finite constant. Furthermore, different from the symmetric case, while $\phi_+^I > \phi_+^{II}$ for all

cases, $\phi_-^I < \phi_-^{II}$ when $\phi_{s,0}$ is small, and $\phi_-^I > \phi_-^{II}$ for higher $\phi_{s,0}$, for the asymmetric mixture with $r > 0$.

In [Figure 2](#), we show the concentration profiles of all charged species along the direction normal to the interface between the supernatant and coacervate phases, for the four bulk systems specified in [Table 1](#). As mentioned earlier, the plane with $x = 0$ is specified by $\phi_{p+}(x = 0) + \phi_{p-}(x = 0) = (\phi_{p+}^I + \phi_{p-}^I + \phi_{p+}^{II} + \phi_{p-}^{II})/2$. From [Figure 2b](#), we see that the polyanion concentration $\phi_{p-}(x)$ increases monotonically from ϕ_{p-}^I to ϕ_{p-}^{II} from the supernatant phase to the coacervate phase for all $\phi_{s,0}$; however, a slight depletion is observed in the polycation concentration profile $\phi_{p+}(x)$ in the interfacial region on the side of the supernatant phase when $\phi_{s,0}$ is small (see the black curve in [Figure 2a](#)). Moreover, going from the supernatant to the coacervate phase, the cation concentration $\phi_+(x)$ decreases monotonically from ϕ_+^I to ϕ_+^{II} at modest values of $\phi_{s,0}$ but exhibits a slight maximum when $\phi_{s,0} = 0$. The anion concentration $\phi_-(x)$, however, exhibits a maximum at the interface for all $\phi_{s,0}$ values examined here, and the value of this maximum becomes less pronounced for larger $\phi_{s,0}$.

With the concentrations of all species, we further calculate the dimensionless net charge density $\phi_e(x) = \phi_{p+}(x) - \phi_{p-}(x) + \phi_+(x) - \phi_-(x)$, as shown in [Figure 3a](#). Due to the charge neutrality in each of the two coexisting phases, we have $\phi_e(x) = 0$ when $x \ll 0$ and $x \gg 0$. An intricate local charge separation behavior, however, is observed in the interfacial region. In general, a positively charged layer forms in the interfacial region on the side of the coacervate phase, and a negatively charged layer forms on the side of the supernatant phase. Moreover, when $\phi_{s,0}$ is small, the charge density profile shows an interesting damped oscillation behavior. Taking $\phi_{s,0} = 0$, as shown by the black curve in [Figure 3a](#), we see that $\phi_e(x)$ exhibits a negative minimum on the right-hand side (i.e., the coacervate side) of the largest positive peak; likewise, a local positive maximum is observed on the left-hand side (i.e., the supernatant side) of the largest negative peak. The degree of charge separation and oscillation in the interfacial region becomes weaker with increasing $\phi_{s,0}$, and charge oscillation disappears at sufficiently large $\phi_{s,0}$. We thus speculate that the formation of the oscillatory electric double-layer structure at small $\phi_{s,0}$ is likely a synergistic effect of electrostatic correlation, chain connectivity, excluded volume interaction, and the charge separation due to asymmetry between oppositely charged species. We note that a similar oscillatory behavior was also observed in other systems, such as concentrated ionic liquid^{96,97} and polyelectrolyte solution confined between neutral surfaces.⁹⁸

In Figure 3b, we show the mean electrostatic potential $\psi(x)$ corresponding to the charge density profiles shown in Figure 3a. For small $\phi_{s,0}$ (e.g., the black curve), when going from the supernatant to the coacervate phase, $\psi(x)$ first decreases slightly and then increases rapidly to Ψ_G , the value of the Galvani potential, after a slight overshoot. When $\phi_{s,0}$ is large, $\psi(x)$ becomes monotonic (e.g., the blue curve).

Finally, with the density and electrostatic potential profiles, we calculate the interfacial tension γ using eq 20; the results are shown in Figure 4 for varying $\phi_{s,0}$ with $r = 1/3$ at $\phi_{p,0} = 0.01$,

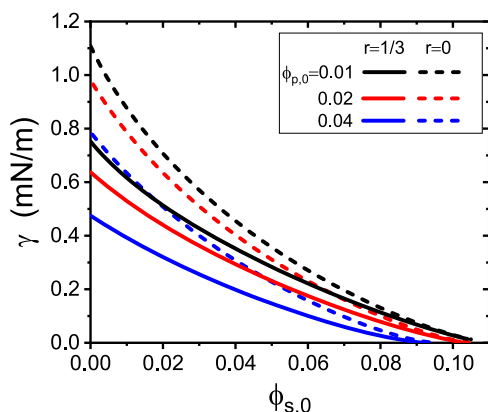


Figure 4. Interfacial tension γ as a function of the added-salt concentration $\phi_{s,0}$ for both asymmetric ($r = 1/3$) and symmetric ($r = 0$) coacervates at several representative values of $\phi_{p,0}$.

0.02, and 0.04. The results for the symmetric coacervate (i.e., $r = 0$) at these three $\phi_{p,0}$ are also included for comparison. Similar to the symmetric case, γ at $r = 1/3$ decreases monotonically with $\phi_{s,0}$ for all the $\phi_{p,0}$ values studied and is smaller for larger $\phi_{p,0}$. At a given $\phi_{s,0}$ and $\phi_{p,0}$, γ for the asymmetric coacervate is less than that of the symmetric coacervate.

3.4. Effects of the Initial Polymer Concentration $\phi_{p,0}$ and Asymmetry Factor r . In this subsection, we examine the effects of the initial polymer concentration $\phi_{p,0}$ and the asymmetry factor r on the interfacial structure and tension of asymmetric coacervates. The concentration profiles for each species (i.e., $\phi_{p+}(x)$, $\phi_{p-}(x)$, $\phi_{+}(x)$, $\phi_{-}(x)$) and the mean electrostatic potential $\psi(x)$ for the different parameter sets are qualitatively similar to Figure 2 and are given in the Supporting Information. Quantitatively, for the coacervate with lower $\phi_{p,0}$, lower $\phi_{s,0}$, and larger r , the depletion of the polycation concentration $\phi_{p+}(x)$ in the interfacial region on the side of the supernatant phase and the charge oscillation in $\phi_c(x)$ become more apparent. In Figure 5, we show the interfacial tension γ as a function of the initial polymer concentration $\phi_{p,0}$ for several representative values of $\phi_{s,0}$ and r . For all cases, γ decreases monotonically with increasing $\phi_{p,0}$; this is mainly due to the fact that the degree of bulk phase separation is gradually weakened with increasing $\phi_{p,0}$ (see Figure S2 in the Supporting Information). Moreover, γ decreases with increasing $\phi_{s,0}$ and r , qualitatively consistent with the results observed in Figure 4.

In Figure 6a, we show the dependence of the interfacial tension γ on the asymmetry factor r , for several representative values of $\phi_{p,0}$ and $\phi_{s,0}$. In general, γ decreases monotonically with increasing r . Interestingly, at $\phi_{s,0} = 0$ (and/or low $\phi_{p,0}$), we find two r regimes, a rapid decrease for small r and a more gradual decrease at larger r . These two regimes become less

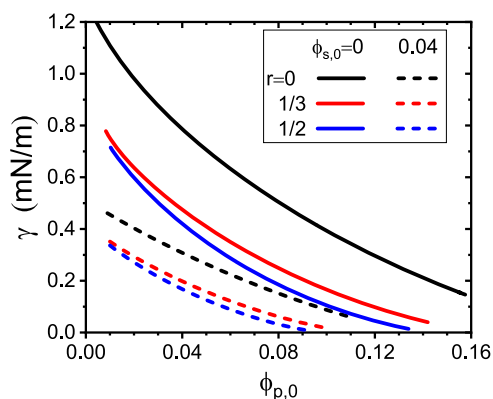


Figure 5. Interfacial tension γ as a function of the initial polymer concentration $\phi_{p,0}$ for both asymmetric ($r = 1/3$ and $r = 1/2$) and symmetric ($r = 0$) coacervates at $\phi_{s,0} = 0$ and 0.04.

apparent for large $\phi_{p,0}$ and $\phi_{s,0}$. The appearance of these two r regimes is closely related to the similar two r regimes observed in the bulk properties as examined in our previous work⁴³ and also reproduced in Figure S2.3 of the Supporting Information for the model in this work.

On the other hand, from eq 20, we recall that γ can be decomposed into one term γ_{ne} that only depends on the concentrations of all species and another term γ_{el} that is directly due to the formation of an electric double layer. We thus calculate these two terms separately to evaluate their relative importance. Taking the system with $\phi_{p,0} = 0.02$ and $\phi_{s,0} = 0$ as an example, in Figure 6b, we show γ_{ne} and γ_{el} separately as a function of the asymmetry factor r . We see that γ_{el} is negative and exhibits a nonmonotonic behavior with r . The negativity of γ_{el} is in accordance with the analysis on the interfacial tension of protein crystals by Sear and Warren⁹⁹ and of the polyelectrolyte and neutral polymer mixture by Nyrkova et al.¹⁰⁰ However, γ is dominated by γ_{ne} , and γ_{el} only plays a minor role. For example, the contribution of γ_{el} to γ is maximal around $r = 0.3$, but even then, its contribution is only about 7%. This suggests that the degree of local charge separation at the interface is modest, which is in turn related to the fact that the two polyion concentrations in the coacervate phase are not very far apart. We thus conclude that the interfacial tension of asymmetric coacervates is mainly determined by the concentration variations of the species in the interfacial region between the two coexisting phases, and not by the electric double layer.

3.5. Relating Interfacial Tension γ to the Degree of Phase Separation S . Since we have shown that the interfacial tension γ is dominated by γ_{ne} that depends only on the concentration of all species, it is tempting to connect γ directly with the concentration of all species in the coexisting phases and explore the relationship between γ and the degree of phase separation. For an uncharged homopolymer solution, it is common to define the polymer concentration difference between the polymer-rich phase and the coexisting polymer-poor phase as the order parameter to quantify the degree of phase separation. For our four-charged component coacervate system, it is thus natural to define an order parameter, $S \equiv \sqrt{\sum_{j=p+,p-,+, -} (\phi_j^{\text{II}} - \phi_j^{\text{I}})^2}$, the Euclidean distance in composition between the coexisting phases, to quantify the degree of phase separation. A similar order parameter has been used in the study of the interfacial tension between a neutral

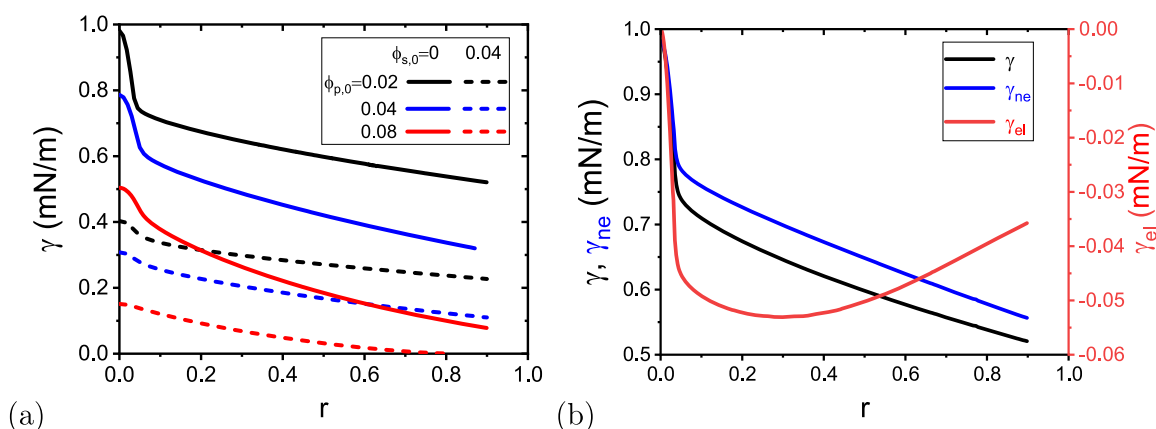


Figure 6. (a) Interfacial tension γ as a function of the asymmetric factor r at several representative values of $\phi_{s,0}$ and $\phi_{p,0}$. (b) Interfacial tension γ and its two components, γ_{ne} and γ_{el} , as a function of r for $\phi_{p,0} = 0.02$ and $\phi_{s,0} = 0$.

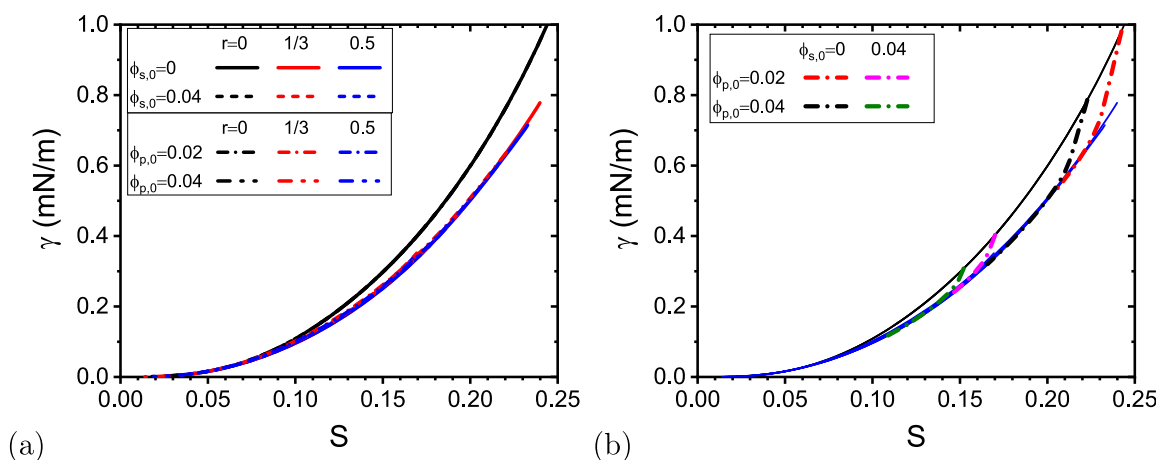


Figure 7. Interfacial tension γ as a function of the degree of phase separation $S \equiv \sqrt{\sum_{j=p+,p-,+,-} (\phi_j^{\text{II}} - \phi_j^{\text{I}})^2}$, for coacervates with (a) varying $\phi_{p,0}$ (the six solid and dash curves in the upper box) and varying $\phi_{s,0}$ (the six dash-dot and dash-dot-dot curves in the lower box) at several values of r and with (b) varying r at different $\phi_{p,0}$ - $\phi_{s,0}$ combinations. In part (b), the two master curves in part (a) are also included.

polymer-rich phase and a polyion-rich phase in an aqueous solution of neutral polymers (dextran) and polyelectrolytes (nongelling fish gelatin).¹⁰¹

In Figures 4 and 5, we have examined the dependence of γ on $\phi_{s,0}$ and $\phi_{p,0}$, respectively, at several values of r ; likewise, based on the bulk concentration of all species in the two coexisting phases, we can also calculate the dependence of S on $\phi_{s,0}$ and $\phi_{p,0}$. By eliminating $\phi_{s,0}$ or $\phi_{p,0}$, we are able to examine the dependence of γ on S . In Figure 7a, we show γ as a function of S for several parameter sets. The black, red, and blue curves correspond to systems with $r = 0, 1/3$, and $1/2$, respectively. Each curve is obtained at fixed r by varying either $\phi_{p,0}$ or $\phi_{s,0}$. For example, the black solid curve with $r = 0$ and $\phi_{s,0} = 0$ is obtained by varying $\phi_{p,0}$, and the red dash-dot curve with $r = 1/3$ and $\phi_{p,0} = 0.02$ is obtained by varying $\phi_{s,0}$ in the calculation. Surprisingly, we find that all curves collapse approximately into two master curves. The upper curve corresponds to the symmetric coacervate with $r = 0$, and the lower one is for the asymmetric coacervate with $r = 1/3$ and 0.5 . Furthermore, at small S , these two curves merge together, and $\gamma \sim S^3$, reflecting the mean-field nature of our theory.

In Figure 7b, we examine the behavior of γ vs S for varying r while keeping $\phi_{p,0}$ and $\phi_{s,0}$ fixed. For reference, the black and blue curves correspond to the two master curves in Figure 7a,

respectively. Taking the black dash-dot curve with $\phi_{p,0} = 0.04$ and $\phi_{s,0} = 0$ as an example, we see that with increasing r from 0, the curve starts from the upper master curve and merges to the lower master curve. The value of r at which the curve merges to the lower master curve is only about $r \approx 0.02$; in other words, all asymmetric coacervates with $r \gtrsim 0.02$ will converge to the lower master curve. A similar phenomenon is also observed for the other three cases.

To understand the mechanism underlying the two master curves, we note that for a given coexisting concentration pair $\{\phi_i^{\text{I}}, \phi_i^{\text{II}}\}$, S is uniquely determined. Likewise, γ is uniquely determined by $\{\phi_i^{\text{I}}, \phi_i^{\text{II}}\}$ and the Galvani potential ψ_G . For the symmetric coacervates, $\psi_G = 0$. From Figure 1a, we see that one value of S corresponds to only one coexisting concentration pair, and thus, different curves of γ vs. S collapse exactly to a single master curve (i.e., the upper master curve in Figure 7), irrespective of how these curves are calculated (i.e., whether they are obtained by varying $\phi_{p,0}$ at fixed $\phi_{s,0}$ or varying $\phi_{s,0}$ at fixed $\phi_{p,0}$ or varying both). For the asymmetric coacervates, while for a given S , there can be different pairs of $\{\phi_i^{\text{I}}, \phi_i^{\text{II}}\}$ and thus different values of γ , our calculation shows that the difference between these γ s is rather small, and thus, different curves of γ vs S collapse approximately to a single master curve (i.e., the lower master

curve in Figure 7). However, the deeper reason underlying the near overlap of the different γ vs S curves for the asymmetric coacervates is unclear and requires further study.

The behavior shown in Figure 7a,b suggests that, once we obtain the main curve(s) of the interfacial tension by measuring a few systems, it is possible to estimate the interfacial tension of other coacervates using only the bulk coexistence concentrations.

4. SUMMARY

In this work, we have developed a simple inhomogeneous mean-field theory to examine the interfacial properties of polyelectrolyte complex coacervates in equilibrium with a polymer-deficient supernatant solution, with a focus on concentration-asymmetric coacervates. Our theory incorporates the translational entropy of all species and models the excluded volume interaction via a local incompressibility condition. The electrostatic correlation is treated within the local density approximation by combining the generalized Debye–Hückel theory⁶⁹ for a simple electrolyte solution with the first-order thermodynamic perturbation theory^{70–72} to include the chain connectivity contribution. The conformation entropy contribution of the polyions is taken into account using the ground-state dominance approximation.^{73,74} Our theory generalizes that by Qin et al.⁶⁰ to allow charge asymmetry and local charge separation that generally exist for asymmetric systems and correctly captures the salt partitioning between the supernatant and coacervate phases observed in experiments and simulation for symmetric mixtures.^{23,38}

For symmetric coacervates, the interfacial tension γ decreases monotonically with increasing the salt concentration in the supernatant phase ϕ_s^I . When ϕ_s^I is close to the critical salt concentration $\phi_{s,c}$, we find that $\gamma \sim |\phi_{s,c} - \phi_s^I|^{3/2}$, in accordance with previous reports by several experimental and theoretical studies.^{56,58–60,63,64} The polyion concentration normal to the interface $\phi_p(x)$ increases monotonically from ϕ_p^I to ϕ_p^{II} , where ϕ_p^I and ϕ_p^{II} are the polycation (or polyanion) concentration in the supernatant phase and coacervate phase, respectively. However, the salt concentration $\phi_s(x)$ exhibits a slight enhancement at the interface when ϕ_s^I is low, and this maximum vanishes with increasing ϕ_s^I .

For concentration-asymmetric coacervates, the charge imbalance between the polycation and polyanion concentrations results in local charge separation in the interfacial region. When the polycations are the major polyion species and polyanions are the minor polyion species, corresponding to the asymmetry factor $r > 0$, the polyanion concentration, $\phi_{p-}(x)$, increases monotonically when going from the supernatant phase to the coacervate phase. However, the polycation concentration, $\phi_{p+}(x)$, exhibits a depletion layer on the supernatant side of the interface, and this depletion is more apparent at large r , low initial polyion concentration $\phi_{p,0}$, and low initial salt concentration $\phi_{s,0}$. Furthermore, the small anion concentration $\phi_-(x)$ exhibits a maximum at the interface for large r , low $\phi_{p,0}$, and low $\phi_{s,0}$, but the cation concentration $\phi_+(x)$ is mostly monotonic. The net charge density, $\phi_e(x) \equiv \sum_{i=p+,p-,+,-} z_i \phi_i(x)$, adopts an intricate double layer structure and develops several oscillations at large r , low $\phi_{p,0}$, and low $\phi_{s,0}$. Correspondingly, the mean electrostatic potential $\psi(x)$ also exhibits a nonmonotonic behavior.

The interfacial tension γ of asymmetric coacervates generally decreases with increasing the asymmetry factor r , the added-salt concentration $\phi_{s,0}$, and the initial polymer concentration

$\phi_{p,0}$. Specifically, at low $\phi_{s,0}$, we find two distinct regimes of the variation in γ with r , an initial rapid decrease followed by a more gradual decrease. We further examine the contribution of the electric double layer to γ and find this contribution to be generally small. Thus, the interfacial tension γ is dominated by the concentration profiles of the species. Moreover, we analyze the relationship between γ and the degree of phase separation, defined by $S \equiv \sqrt{\sum_{j=p+,p-,+,-} (\phi_j^{II} - \phi_j^I)^2}$, and find that the data collapse to two master curves, one for symmetric coacervates and one for asymmetric coacervates, and that at small S , the two curves converge to one described by the mean-field scaling behavior $\gamma \sim S^3$.

The magnitude of the interfacial tension for all the cases studied is generally small, on the order of 1 mN/m or less. The smallness of the interfacial tension may be responsible for the fact that the coacervate phase often exists as many droplets.^{20,21} Here, we have considered the case where the coacervate phase is macroscopic, and thus, charge neutrality is naturally satisfied. This assumption should be valid for systems with weak and modest asymmetry or moderate salt concentration. For strong asymmetry and low salt concentration, the mixture may not undergo a macroscopic phase separation but instead form soluble finite-sized polyion complex clusters with an optimal amount of net charge.^{102–104} For example, Oskolkov and Potemkin predicted theoretically the formation of finite-sized charged clusters of lamellar, cylindrical, and spherical shapes for weakly concentration-asymmetric mixtures of polycation and polyanion solutions.¹⁰⁴ The structure of such a charged cluster will likely affect the bulk phase behaviors of asymmetric coacervates. In addition, the supernatant phase is not a featureless ultradilute solution of isolated polyions but is also made of finite-sized polyion complex clusters. Even for symmetric coacervates, Delaney and Fredrickson have shown that far from the critical point, the dilute phase consists primarily of polycation–polyanion dimers.³⁶ The theory by Muthukumar also assumes the dimer to be the smallest structural units in coacervation.^{39,81,105} Rumyantsev and Potemkin pointed out the formation of neutral globules at concentrations well below the dilute branch of the phase boundary.³⁵ Assuming that these clusters on average are spherical in shape, the theory developed in this work can be easily extended to study these finite-sized clusters; such an effort will be pursued in future work.

■ ASSOCIATED CONTENT

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.macromol.1c01809>.

Expressions for the chemical potential of each species and the osmotic pressure of the bulk solution; bulk concentration of the different species and Galvani potential at coexistence, as a function of the added-salt concentration, initial polymer concentration, and asymmetry factor; and concentration profiles of all species, net charge density, and mean electrostatic potential at the interface of asymmetric coacervates for different $\phi_{p,0}$ and r (PDF)

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Notes

The authors declare no competing financial interest.

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